

Practice Problems:
AC Circuit Analysis
Introduction to Electric Circuits
Supplemental Instruction

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This document is not a substitute for a textbook or classroom instruction. It may contain errors and is intended to be used only to the extent that it is helpful to supplement students' learning experience.

1 Sinusoidal Function Combinations

Given $x(t) = 4 \cos(3t - 20^\circ)$ and $y(t) = 8 \cos(3t + 60^\circ)$, simplify $x(t)y(t)$ to a single cos function.

Solution

Let's convert both functions to phasors, taking $\omega = 3$. $\mathbf{X} = 4\angle -20^\circ$ and $\mathbf{Y} = 8\angle 60^\circ$. Therefore, the product is $4 \cdot 8\angle(-20 + 60)^\circ = 32\angle 40^\circ$. Converting back to a cos function, $x(t)y(t) = 32 \cos(3t + 40^\circ)$.

2 Phasors

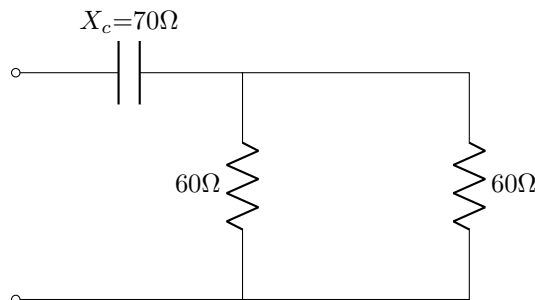
For the voltage $\mathbf{V} = 5\angle 30^\circ$ V, find $V(t)$ as a cosine function and find $V(1.893\text{s})$. Assume a frequency $f = 100$ Hz.

Solution

The angular frequency is $\omega = 2\pi f = 2\pi \cdot 100\text{Hz} = 200\pi$ rad/s. Therefore, $V(t) = 5 \cos(200\pi t + 30^\circ)$ V. When finding the voltage at a specific time, it is important to be careful about the fact that ω is in units of rad/s, but the phase angle is in degrees: $V(2\text{s}) = 5 \cos(200\pi \text{ rad} \cdot 1.893 + 30^\circ)\text{V} = 5 \cos(378.6\pi + \pi/6)\text{V} = -3.716$ V.

3 Impedance Combinations

Find the total impedance in both polar and rectangular form.

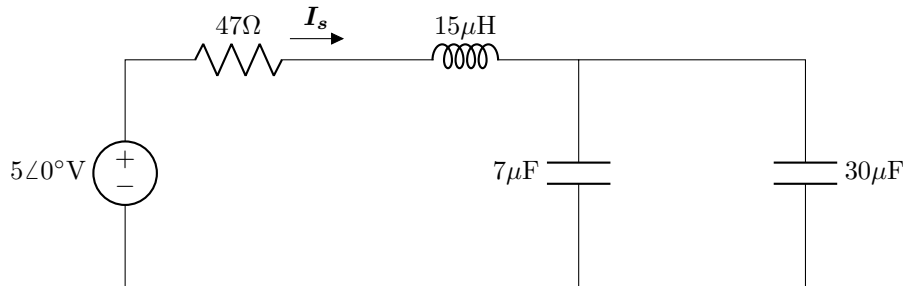


Solution

Recall that capacitor impedance is always negative. The total impedance is $\mathbf{Z}_{tot} = (60\Omega || 60\Omega) - j70\Omega = 30 - j70\Omega$. Converting to polar form, $Z_m = \sqrt{30^2 + 70^2} = 76.161$, and $\theta = \arctan(\frac{-70}{30}) = -66.8^\circ$, so $\mathbf{Z}_{tot} = 76.16\angle -66.8^\circ\Omega$.

4 Impedance Combinations, Ohm's Law

Find the total impedance in polar form, and the total current $\mathbf{I}_s$. Assume $f = 60\text{ Hz}$.



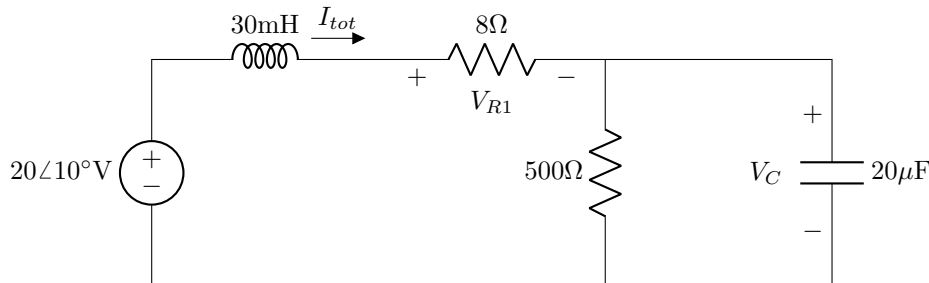
Solution

We can combine the capacitors in parallel by adding them to get $37\mu\text{F}$. Now, let's find the impedances of the reactive components. $\mathbf{Z}_L = j \cdot 2\pi 60 \cdot 15\mu\text{H} = 0.005655j\Omega$, and $\mathbf{Z}_C = \frac{1}{j \cdot 2\pi 60 \cdot 37\mu\text{F}} = -71.691\Omega$. Therefore, the total impedance is $\mathbf{Z}_{tot} = 47 + 0.005655j - 71.691j = 47 - 71.69j\Omega$. We can convert this to polar form to obtain $\mathbf{Z}_{tot} = 85.72\angle -56.75^\circ\Omega$.

To find the total current, we will apply Ohm's Law: $\mathbf{I}_s = \frac{\mathbf{V}}{\mathbf{Z}_{tot}} = \frac{5\angle 0^\circ\text{V}}{85.72\angle -56.75^\circ\Omega} = 58.3\angle (0^\circ - (-56.75^\circ)) = 58.3\angle 56.75^\circ\text{ mA}$.

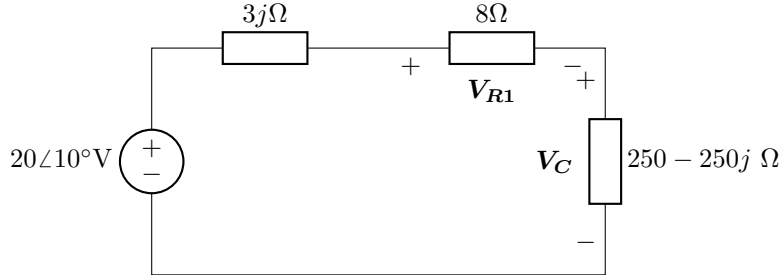
5 Voltage Division in AC

Find as phasors (a) the total current $\mathbf{I}_{tot}$, (b) V_{R1} , and (c) V_C . Assume $\omega = 100$.



Solution

The total impedance is $Z_{tot} = j\omega 30\text{mH} + 8 + (500 \parallel \frac{1}{j\omega 20\mu\text{F}}) = j100 \cdot 30 \cdot 10^{-3} + 8 + \frac{1}{\frac{1}{500} + j100 \cdot 20 \cdot 10^{-6}} = 258 - 247j \Omega$. Therefore, the total current is $I_{tot} = \frac{20\angle 10^\circ \text{V}}{258 - 247j \Omega} = \frac{20\angle 10^\circ \text{V}}{357.174\angle -43.7522^\circ \Omega} = 56\angle 53.75^\circ \text{ mA}$. We can find V_{R1} and V_C using voltage division. We have a circuit with three equivalent impedances in series:



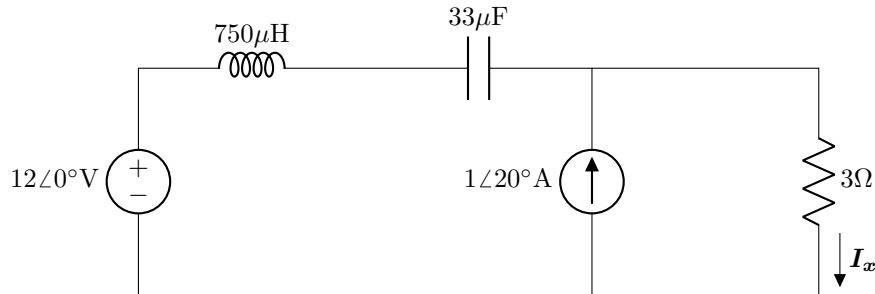
We can find the voltages as follows:

$$V_{R1} = 20\angle 10^\circ \cdot \frac{8}{3j + 8 + 250 - 250j} = 448\angle 53.8^\circ \text{ mV}$$

$$V_C = 20\angle 10^\circ \cdot \frac{250 - 250j}{3j + 8 + 250 - 250j} = 19.8\angle 8.75^\circ \text{ V}$$

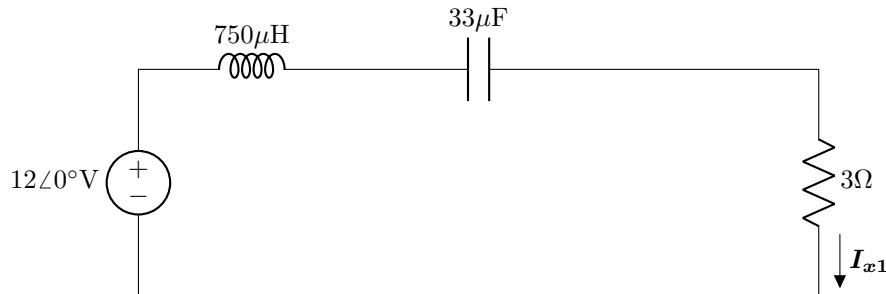
6 Superposition

Find I_x . Assume $\omega = 2000$.



Solution

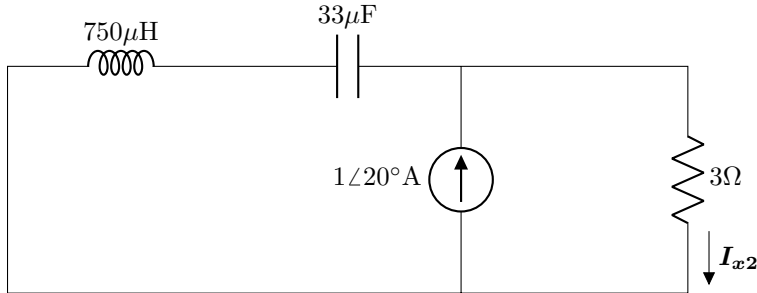
We will first turn off the current source:



Using Ohm's Law,

$$\mathbf{I}_{x1} = \frac{12}{j2000 \cdot 750\mu\text{H} + \frac{1}{j2000 \cdot 33\mu\text{F}} + 3\Omega} = 0.1843 + 0.8385j \text{ A}$$

Next, we turn off the voltage source:



The combined impedance of the inductor and capacitor is $-13.6515j \Omega$. Using current division,

$$\mathbf{I}_{x2} = (\cos(20^\circ) + j \sin(20^\circ)) \cdot \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{-13.6515j}} = 0.9681 + 0.1293j \text{ A}$$

Therefore, the total current is:

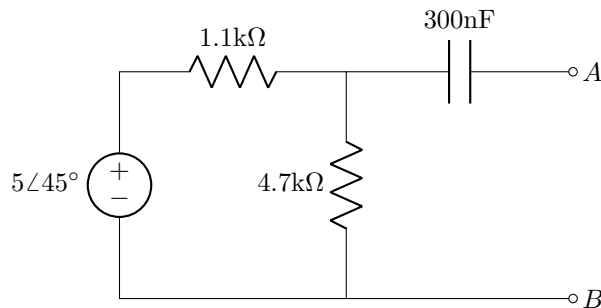
$$\mathbf{I}_x = 0.1843 + 0.8385j + 0.9681 + 0.1293j = 1.1524 + 0.9678j \text{ A}$$

This can also be converted to polar form:

$$\mathbf{I}_x = 1.505 \angle 40.023^\circ \text{ A}$$

7 Thevenin

Find the thevenin equivalent between points A and B . Assume a frequency of 1000 Hz.



Solution

For the open-circuit voltage, no current flows through the capacitor, so the thevenin phasor voltage is $\mathbf{V}_{th} = 5 \angle 45^\circ \cdot \frac{4.7}{1.1 + 4.7} = 4.0517 \angle 45^\circ \text{ V}$. For the thevenin impedance, the voltage source is turned off (shorted), so the thevenin impedance is $\mathbf{Z}_{th} = \frac{1}{j2000\pi \cdot 300\text{nF}} + (1.1\text{k} \parallel 4.7\text{k}) = 891.379 - 530.516j \Omega = 1.037 \angle -30.76^\circ \text{ k}\Omega$.

References

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- M. Iordache. 2024. EEGR 2053 Electric circuits [PowerPoint slides]. Available: <https://mviordache.name/EEGR2053/index.html>
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